

Zachary James McNulty

CONTACT INFORMATION	1004 Evans Hall Berkeley, CA 94720	zachary_mcnulty@berkeley.edu https://math.berkeley.edu/~zmcnulty/
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EDUCATION **University of California, Berkeley**

Ph.D. Candidate in Applied Mathematics

- Advisor: Steve Evans
- Interests: Combinatorial stochastic processes and random trees

Graduate Certificate in Applied Data Science

University of Washington, Seattle

B.S. Applied Mathematics (3.98/4, summa cum laude), January 2020

- Concentration in Discrete Math and Algorithms

Relevant Coursework: Statistical learning theory, random matrices, stochastic processes, and advanced probability. A full course list is available on my website

PROGRAMMING LANGUAGES	Proficient: Python, Bash/Unix, $\text{\LaTeX}$, git Familiar: C/C++, SQL, Java, MATLAB, GitHub: zackmcnulty
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EXPERIENCE **Five Rings Capital**

Quantitative Research Intern	Summer 2024
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- Completed a data analysis and statistical modeling project contributing to the firm's quantitative trading strategies.
- Received instruction on a wide range of financial concepts.
- Utilized Python data analysis and visualization libraries

Evans Research Group

Graduate Student Researcher	2022+
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- Extending graphon theory of dense graphs to other combinatorial stochastic processes.
- Characterizing Doob-Martin compactification of the uniform d -ary tree Markov process produced by Marckert's algorithm.
- Developing a scaling limit for the NNI tree-exchange process as a Markov chain on the space of $\mathbb{R}$ -trees

UW Shea-Brown Lab

Undergraduate Training Program in Computational Neuroscience	2018-2020
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- Studied how RNNs extract low-dimensional representations of latent variables underlying predictive learning tasks
- Implemented neural networks and a variety of dimensionality reduction techniques in Tensorflow/Keras

Center for Reproducible Biomedical Modeling

Summer Undergraduate Internship	Summer 2018
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- Improved/documented Systems Biology software (Tellurium)
- Developed software for potential DREAM Challenge related to inferring causality in randomized gene regulatory networks

	UW Research Computing Club	
	Summer Undergraduate Leader	2018-2020
	◦ Shared my experience with slurm schedulers, utilizing GPUs, and interacting with remote servers for HPC ◦ Received training in parallel-computing, pyMPI, and OpenMP	
SCHOLARSHIPS, GRANTS, AND AWARDS	UC Berkeley Mathematics Summer Grant	2022-2024
	University of Washington Honors Undergraduate Scholar	2016-2018
	NIKE Frank Rudy Scholarship	2016-2020
	National Scholar Foundation Peer Tutor	2016-2018
PRESENTATIONS	<i>Total Variation Bounds in Poisson Approximation Using Size-Biased Coupling</i> , Student Probability Seminar, UC Berkeley (Spring 2023)	
	<i>A Spatial Markov Property for the Continuum Gaussian Free Field</i> , Student Probability Seminar, UC Berkeley (Spring 2022)	
	<i>Applications of Spectral Bounds on Markov Chain Mixing Times to Projected and Product Chains</i> , Student Probability Seminar, UC Berkeley (Fall 2021)	
	<i>Inferring Low-Dimensional Latent Structures with Recurrent Neural Networks</i> , Department of Mathematics, University of Washington. (June 2019)	
CERTIFICATIONS AND WORKSHOPS	Graduate Certificate in Applied Data Science, UCB 2024	
	Random Matrix Theory, OSU Summer School 2023	
	Concentration in High-Dimensional Probability and Geometry, MSRI Summer School 2023	
	TCS Summer School on Optimal Markov Chain Mixing, UCSB	
	Undergraduate Training Program in Computational Neuroscience, University of Washington Computational Neuroscience Center	
TEACHING AND ORGANIZATIONAL	Probability Student Seminar Organizer	
	UC Berkeley Department of Mathematics	
	◦ Superconcentration of Measure (Fall 2023)	
	Directed Reading Program	
	UC Berkeley Department of Mathematics	
	◦ Martingales and Measure-theoretic probability (Fall 2024) ◦ Time Series Analysis by State Space Methods (Fall 2023)	
	Graduate Student Instructor	
	UC Berkeley Department of Mathematics	
	◦ Integral Calculus (Fall 2020 - Fall 2021) ◦ Linear Algebra and Differential Equations (Spring/Summer 2022) ◦ Multivariable Calculus (Fall 2022, Summer 2023) ◦ Discrete Math (Spring 2023 - Fall 2024) ◦ Numerical Analysis for Data Science and Statistics (Fall 2023) ◦ Certificate in Teaching and Higher Learning (in-progress)	

REFERENCES

Prof. Steve Evans, UC Berkeley (evans@stat.berkeley.edu)

Prof. Eric Shea-Brown, University of Washington (etsb@amath.washington.edu)

Prof. Nathan Kutz, University of Washington (kutz@uw.edu)

VOLUNTEERING AND OUTREACH

Seattle Animal Shelter and Rabbit Ears Rescue

- Helped feed, clean, and socialize animals to prepare them for adoptions as well as help traumatized animals return to normalcy
- Volunteered at adoption events to help find animals suitable future homes and educate the local community on proper animal care

Tent City Collective

- Worked with local communities and legislators to advocate housing reform
- Volunteered for Tent City 3 and helped organize their stay at UW, the first public university in the nation to agree to do so

Camp Kesem

- Helped run Camp Kesem, a summer camp for children whose parents are fighting cancer, helping them to process their emotions in a supportive environment.

Engineering Outreach

- Helped run UW's annual Engineering Discovery Days outreach event
- Ran events at local high schools to help foster curiosity and passion in STEM